



Temperature_C	1	0.009620243	0.03803978	-0.005094283	-0.008846062	-0.004703574
Vibration_Hz	0.009620243	1	0.01185593	-0.01486327	0.01215157	0.008984324
Power_Consumption_kW	0.03803978	0.01185593	1	-0.04005268	0.003789851	-0.2526589
Predictive_Maintenance_Score	-0.005094283	-0.01486327	-0.04005268	1	-0.01107441	-0.02219012
Error_Rate_%	-0.008846062	0.01215157	0.003789851	-0.01107441	1	0.06175295
Anomaly_Score	-0.004703574	0.008984324	-0.2526589	-0.02219012	0.06175295	1

Temperature_C

Vibration_Hz

Power_Consumption_kW

Predictive_Maintenance_Score

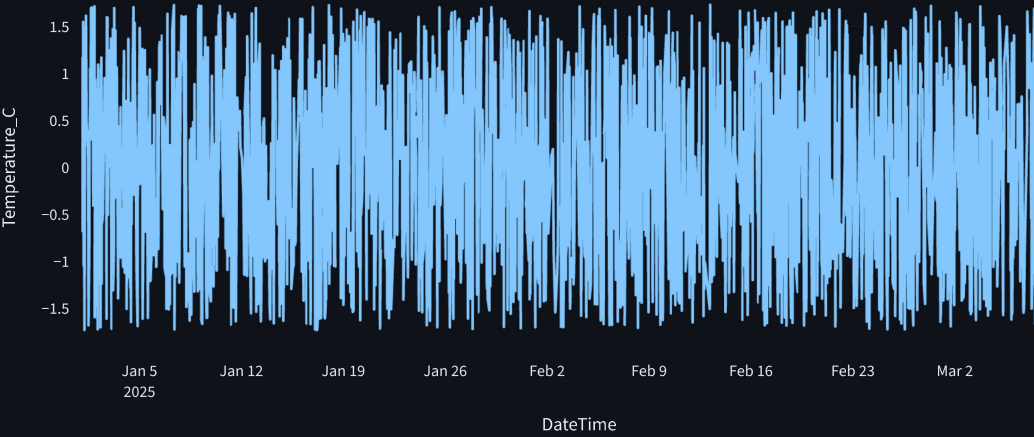
Error_Rate_%

Anomaly_Score

Machine Analysis Dashboard

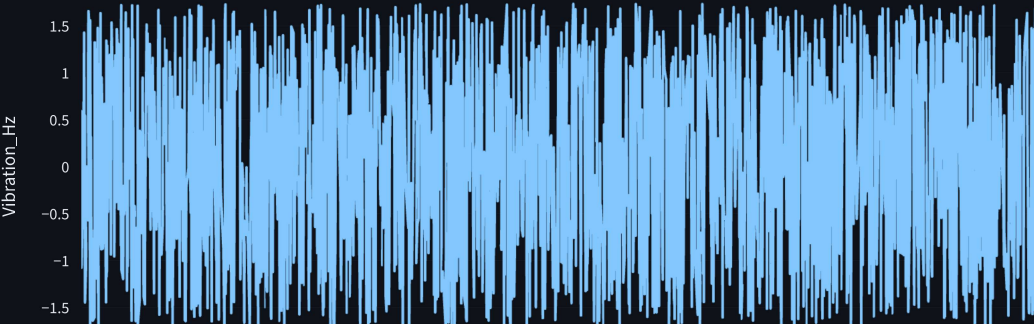
Temperature Trend

Machine Temperature



Vibration Trend

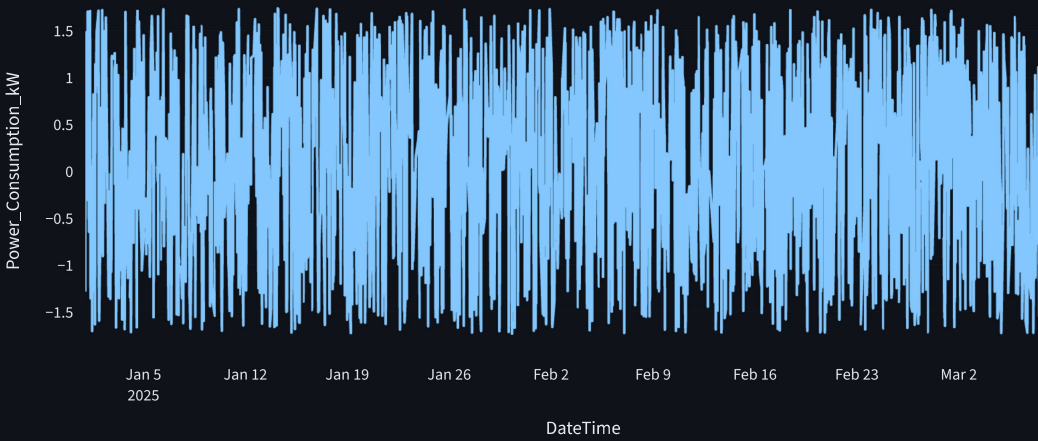
Machine Vibration



Jan 5 2025 Jan 12 Jan 19 Jan 26 Feb 2 Feb 9 Feb 16 Feb 23 Mar 2
DateTime

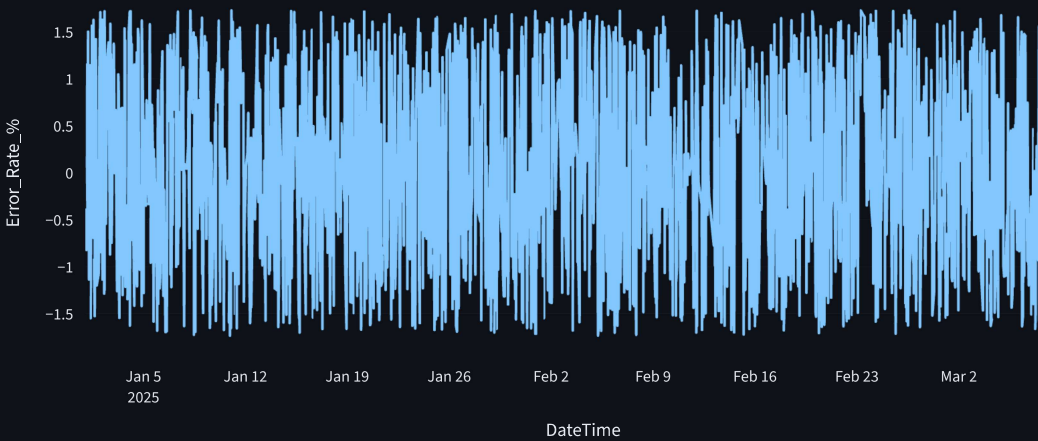
Power Consumption

Power Consumption



Error Rate

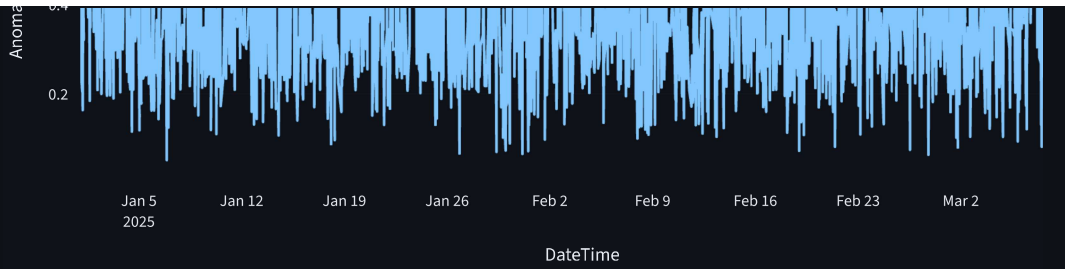
Machine Error Rate



Anomaly Score Trend

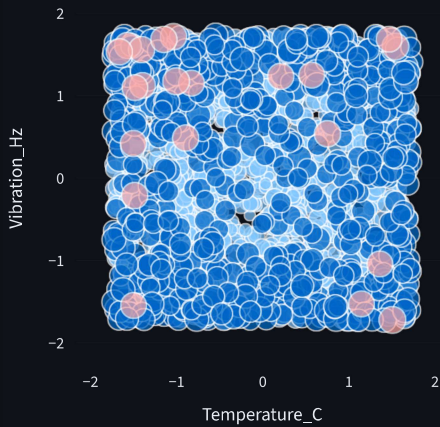
Anomaly Score



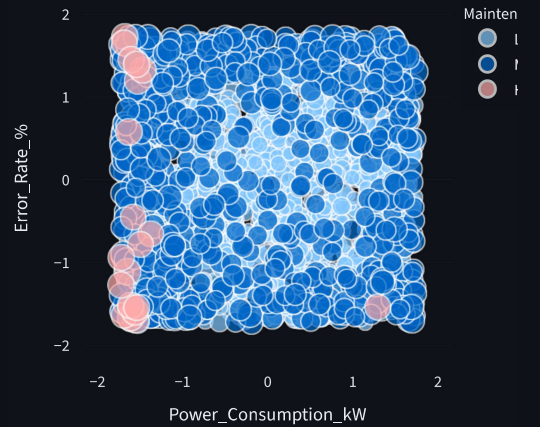


Sensor Comparison

Temperature vs Vibration

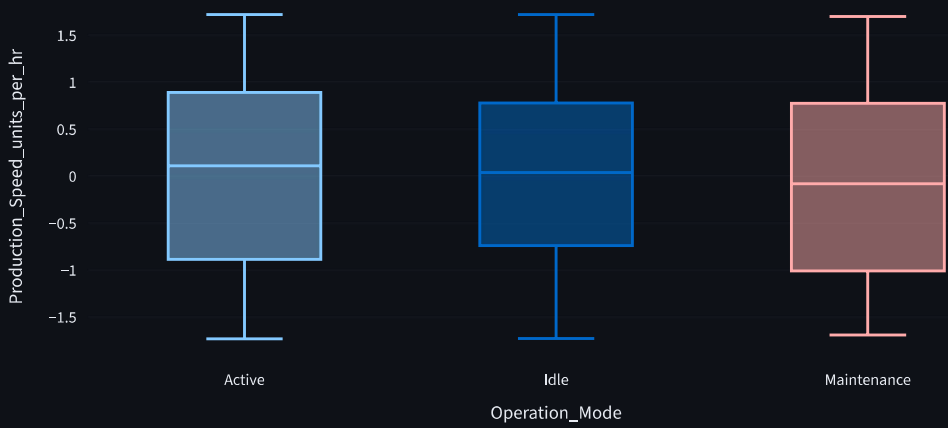


Power vs Error Rate



Production Analysis

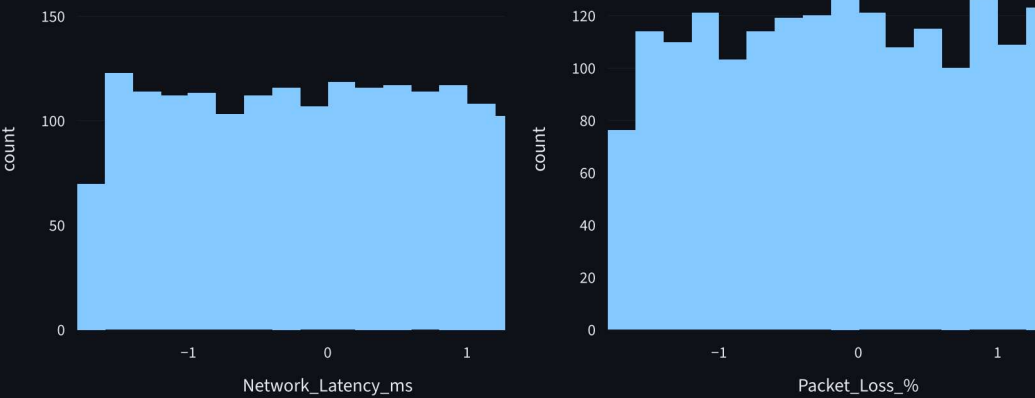
Production Speed by Operation Mode



6G Network Performance

Network Latency

Packet Loss



Machine Performance Summary

	Machine_ID	Temperature_C	Vibration_Hz	Power_Consumption_kW	Error_Rate_%	Anomaly_Score	Predictive_Ma
0	3	-0.0364	0.0381	0.0679	0.022	0.3672	



Maintenance Alert Panel

⚠️ 1 High-Risk Machines Detected

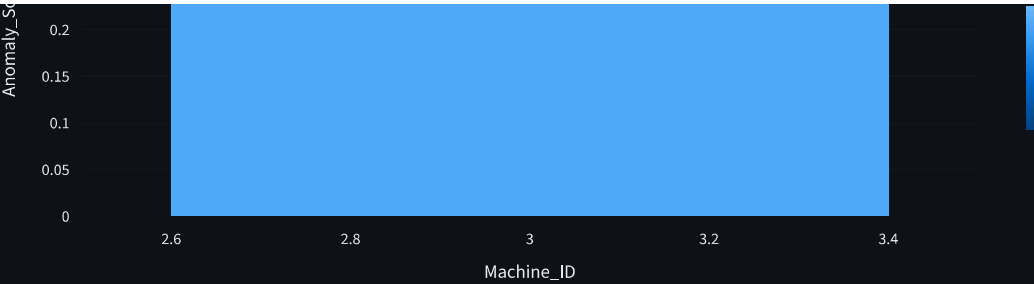
	Machine_ID	DateTime	Temperature_C	Vibration_Hz	Power_Consumption_kW	Error_Rate_%	Predictive
4877	3	2025-02-01 15:35:00	1.5398	1.5947	-1.6293	0.5761	
4709	3	2025-01-26 23:52:00	-1.0053	1.1725	-1.6465	-1.1124	
4136	3	2025-01-07 06:37:00	1.5045	-1.7212	-1.5249	1.1951	
4161	3	2025-01-08 05:25:00	-1.4868	-0.2128	-1.6771	1.7258	
5848	3	2025-03-06 16:35:00	-1.5014	0.418	-1.7199	-0.9394	
5869	3	2025-03-07 11:57:00	-1.6648	1.531	-1.5396	1.3923	
4168	3	2025-01-08 13:34:00	-1.568	1.6067	-1.5506	-1.692	
4608	3	2025-01-23 23:30:00	-1.0363	1.7149	-1.5567	-1.5974	
4504	3	2025-01-19 22:47:00	-0.8948	0.4844	-1.6098	-1.6666	
4771	3	2025-01-28 21:59:00	-1.4065	1.5572	-1.4649	1.3243	



Top 10 Critical Machines

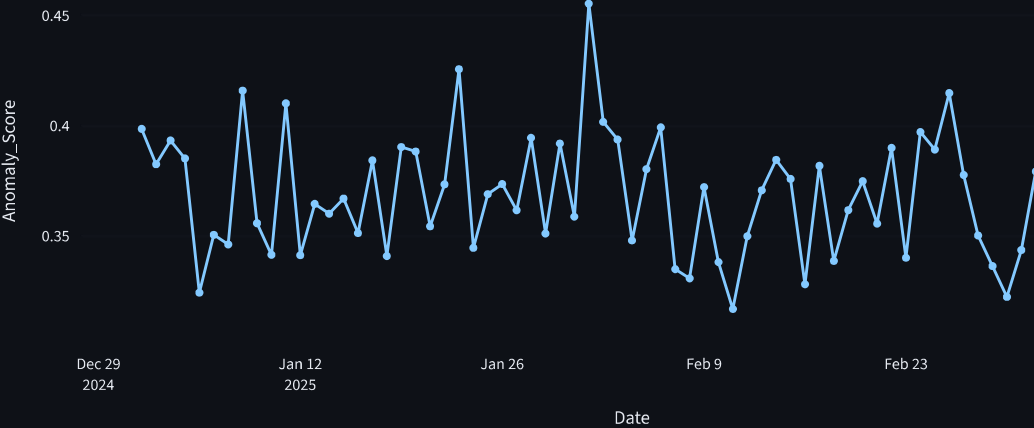
Top 10 High-Risk Machines



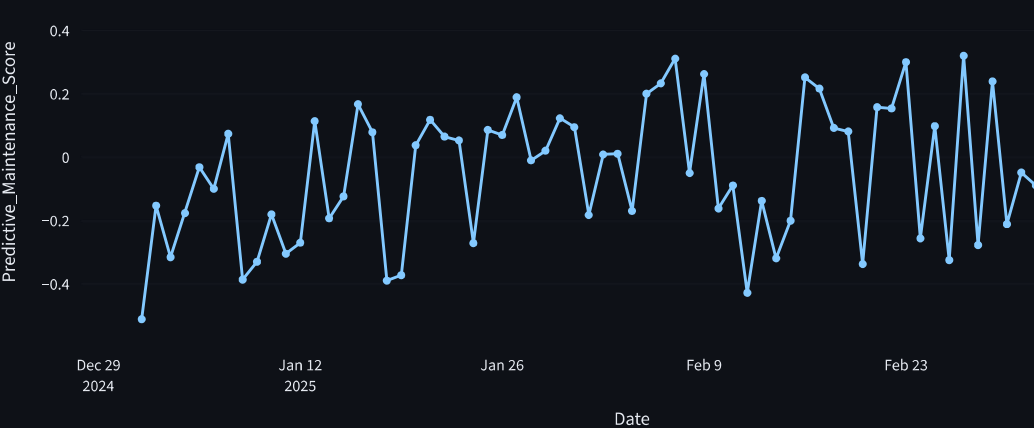


Historical Risk Analysis

Average Anomaly Score Over Time

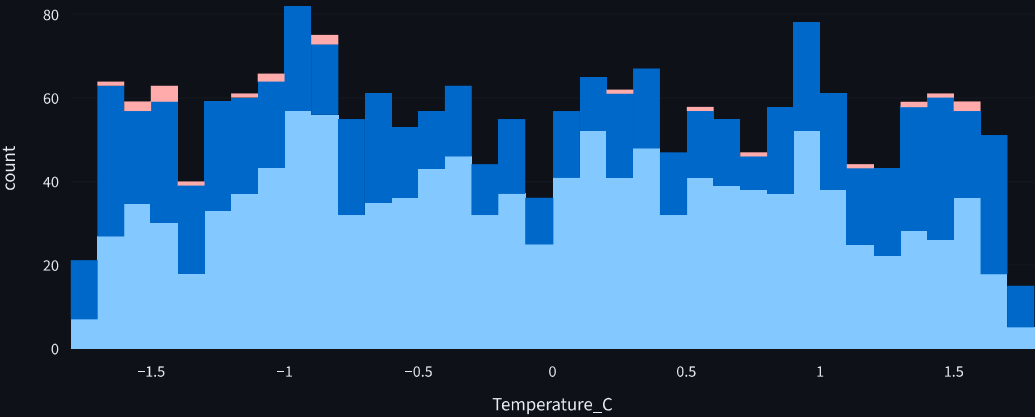


Maintenance Score Trend

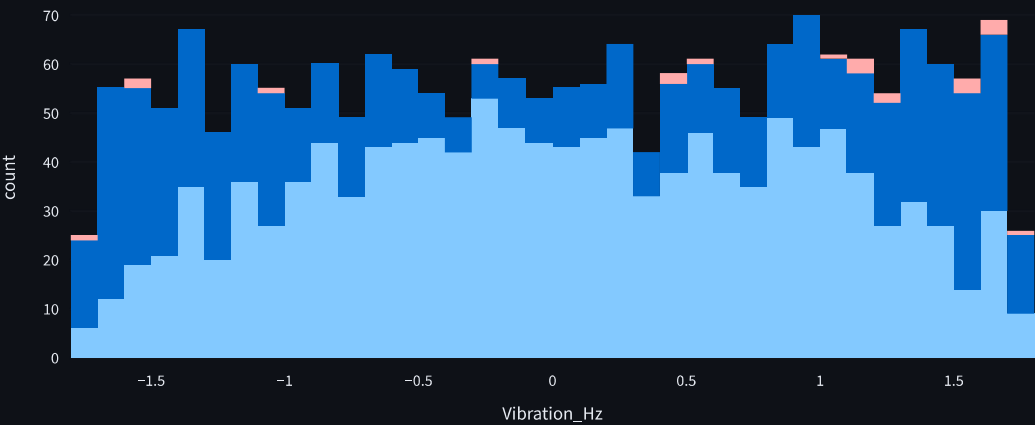


Temperature Distribution

Temperature Distribution



Vibration Distribution



Download Filtered Dataset

Download CSV



Executive Summary

Project Insights

- Total Machines: 1
- High Risk Machines: 1
- Medium Risk Machines: 1
- Low Risk Machines: 1

Recommendations

- Prioritize inspection of High-Risk machines.
- Monitor increasing anomaly scores to prevent unexpected failures.

- Track temperature and vibration jointly for early warning.
- Review machines with declining predictive maintenance scores.
- Use anomaly trends to schedule preventive maintenance and reduce downtime.



Predictive Maintenance Dashboard Loaded Successfully!